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Explainable AI in agriculture: review of applications, methodologies, and future directions

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Abstract

Agriculture forms the backbone of the global economy, facing mounting pressure from population growth and resource constraints. The sector increasingly relies on data-driven technologies to enhance productivity while reducing environmental impact. Agriculture is being revolutionized by Artificial Intelligence (AI), which is enhancing pesticide application, weed control, and irrigation management. Deep Learning techniques that have demonstrated predictive power include Generative Adversarial Networks, Recurrent Neural Networks, and Convolutional Neural Networks. Their opacity and intricacy, however, make practical use difficult. In agricultural settings, Explainable AI (XAI) enables informed decisions by providing transparency without compromising performance. This comprehensive review analyzes peer-reviewed publications from 2020 onwards, categorizing XAI techniques and their applications in agriculture. The starting point of 2020 was deliberately chosen to capture the most recent advancements, as this period marks a phase of rapid growth and wider adoption of XAI within agricultural AI applications, making it particularly relevant for reflecting state-of-the-art developments. This review identifies significant challenges, current research trends, methodological approaches, and evaluate the efficacy of various explainability methods, including LIME, SHAP, Grad-CAM, and rule-based models. The analysis examines key domains including crop-weed discrimination, plant disease detection, precision farming techniques, yield forecasting, and soil quality assessment. The integration of XAI methodologies in precision agriculture presents promising opportunities to address pressing challenges related to resource optimization, climate adaptation, and global food security. This review also provides a structured framework for future research directions and practical implementation guidelines to enhance the interpretability, trustworthiness, and adoption of AI-powered agricultural systems among farmers, agronomists, and policymakers.

Nomenclature

AI	Artificial Intelligence
AIRS	Aerial Imagery of Roof Segmentation Dataset, a public dataset that aims at benchmarking the algorithms of roof segmentation from very-high-resolution aerial imagery
DCGAN	Deep Convolutional Generative Adversarial Network
DCNN	Deep Convolutional Neural Network
DL	Deep Learning
FRCNN	Faster R-CNN
GAN	Generative Adversarial Network
Grad-CAM	Gradient-weighted Class Activation Mapping

LIME	Local Interpretable Model-agnostic Explanations
ML	Machine Learning
OPPD	Open Plant Phenotyping Dataset, a public dataset for visual recognition tasks on images of plant seedlings
PSD	Plant Seedlings Dataset, comprises 960 unique plant images from 12 species at various growth stages, arranged in annotated RGB format with a resolution of 10 pixels per mm.
SHAP	Shapley Additive Explanations
XAI	Explainable AI
YOLO	You Only Look Once

1. Introduction

Agriculture—the foundation of food, fiber, and commodity production through soil cultivation—confronts numerous critical challenges in the contemporary era. Disease outbreaks, diminishing land and water resources, climate change impacts, and persistent pest infestations collectively threaten global agricultural systems. With projections indicating the need for a 70% increase in agricultural output to meet rising demands, the sector stands at a crucial juncture. The implementation of intelligent farming methodologies has become imperative to achieve sustainability goals, strengthen food security frameworks, enhance production efficiency, and significantly reduce environmental footprints. These sophisticated approaches represent a necessary evolution in agricultural practice, balancing productivity requirements with ecological responsibility to address the complex challenges of feeding a growing global population while preserving natural resources for future generations.

1.1. Overview of AI in agriculture

The field of Artificial Intelligence (AI) within computer science is growing at a remarkable pace, and its influence is strongly felt in agriculture—a sector that employs nearly one-third (30.7%) of the global workforce and covers 2781 million hectares of cultivated land. Across this vast landscape, AI has emerged as a powerful ally, offering solutions to some of the most pressing challenges in farming. From disease identification and pest management to precision chemical application, drainage and irrigation optimization, weed control, and crop yield forecasting [1], AI has consistently proven its value. These systems have shown exceptional accuracy and reliability in managing complex agricultural scenarios, with an unmatched ability to detect and analyze the subtle details that define diverse farming conditions [2]. One striking example lies in precise spraying, where weed detection plays a crucial role in reducing environmental damage, cutting costs, and minimizing chemical use. This is an area where AI—particularly through ML and DL—has made significant progress. Earlier, weed species in agricultural fields were identified using image processing algorithms that extracted textural and morphological features. However, these methods required manual feature extraction and often struggled with challenges such as varying field conditions, changing illumination, and occlusion. DL models have since revolutionized this domain, offering far greater accuracy and adaptability than traditional machine learning approaches. Yet, challenges remain, including limited datasets, environmental variability, and class imbalance. Overcoming these hurdles and building scalable, dependable AI-powered precision spraying systems is critical for ensuring sustainable weed management [3].

AI plays a crucial role in irrigation management, integrating soil moisture sensor data, crop growth stages, and weather forecasts to optimize water distribution. ML models predict irrigation needs, reducing water wastage and ensuring adequate hydration, supporting sustainable farming practices, especially in regions facing water scarcity.

AI's transformative potential is also evident in the realm of crop yield prediction and sustainable farming practices. By harnessing vast amounts of data, AI empowers farmers to make informed decisions, optimize resources, and reduce production costs. Tools such as satellite imaging and remote sensing supply extensive information on crop health, soil quality, and weather conditions. ML models, trained on historical yield data and environmental factors, then use this knowledge to forecast future harvests. With the ability to monitor numerous yield-related variables, process large datasets at speed, and generalize across crops and geographies, AI provides timely insights and early detection of potential risks. These predictive capabilities not only enhance output but also lower costs and strengthen sustainability, paving the way for more resilient and secure farming systems [4, 5].

Beyond precision spraying and irrigation management, AI is also being integrated into broader agricultural systems to enhance productivity and sustainability. AI-driven solutions support tasks such as

automated planting schedules, improved weather forecasting, fertilizer optimization, soil assessment, and plant spacing, achieving predictive accuracies exceeding 90% [6]. Intelligent diagnostic expert systems and regression-based models—including Random Forest, Support Vector Machines (SVM), and Multiple Polynomial Regression have shown strong performance in yield prediction and agronomic decision-making [7]. Moreover, AI-powered decision support systems assist with seed timing, crop quality assessment, microclimatic forecasting, and soil fertility evaluation, while real-time monitoring tools diagnose plant stress, nutrient imbalance, and disease outbreaks to strengthen precision agriculture practices [8]. UAVs equipped with computer vision and DL algorithms enable targeted agrochemical applications, reducing chemical input wastage and mitigating environmental risks [9]. The integration of Internet of Things (IoT) technologies, including wireless sensor networks (WSNs), cloud computing platforms, and RFID, has further advanced mechanized farming, greenhouse automation, and real-time monitoring in precision agriculture, improving operational efficiency, resource optimization, and reducing labor dependency [10].

Figure 1 depicts an overview diagram of the review framework.

1.2. The focus of agricultural ai on explainability

ML and DL algorithms in agriculture are beneficial but often lack transparency, leading to concerns about safety, fairness, and ethics. Explainable Artificial Intelligence (XAI) aims to address these issues by making AI models more transparent, fostering better understanding, sound judgment, and increased confidence in AI systems. This has led to a rise in XAI's importance across various scientific fields.

XAI techniques can be categorized into three main groups based on methodological generality, explanation scope, and model selection approach. For model selection, practitioners can choose between model-based methods (simpler but requiring more sophisticated algorithms) and post-hoc methods. Both model-specific and model-agnostic XAI approaches offer the flexibility to generate either local explanations (for individual predictions) or global explanations (for overall model behavior) [11, 12].

Figure 2 illustrates different concepts of XAI as applied in the agricultural sector.

1.3. Motivation of review paper

Despite its potential, there is a noticeable gap in the literature regarding the application of XAI in agriculture. There is a lack of comprehensive studies summarizing the various XAI techniques employed in agricultural tasks, as well as limited research on how XAI can enhance specific applications, such as weed detection, disease classification, and yield prediction. This review aims to bridge this gap by providing a structured analysis of XAI applications in agriculture, highlighting key methodologies, challenges, and future research directions. By addressing the interpretability limitations of DL models, this study seeks to advance the integration of XAI-driven solutions in precision agriculture for improved efficiency, sustainability, and decision-making.

1.4. Objectives of review paper

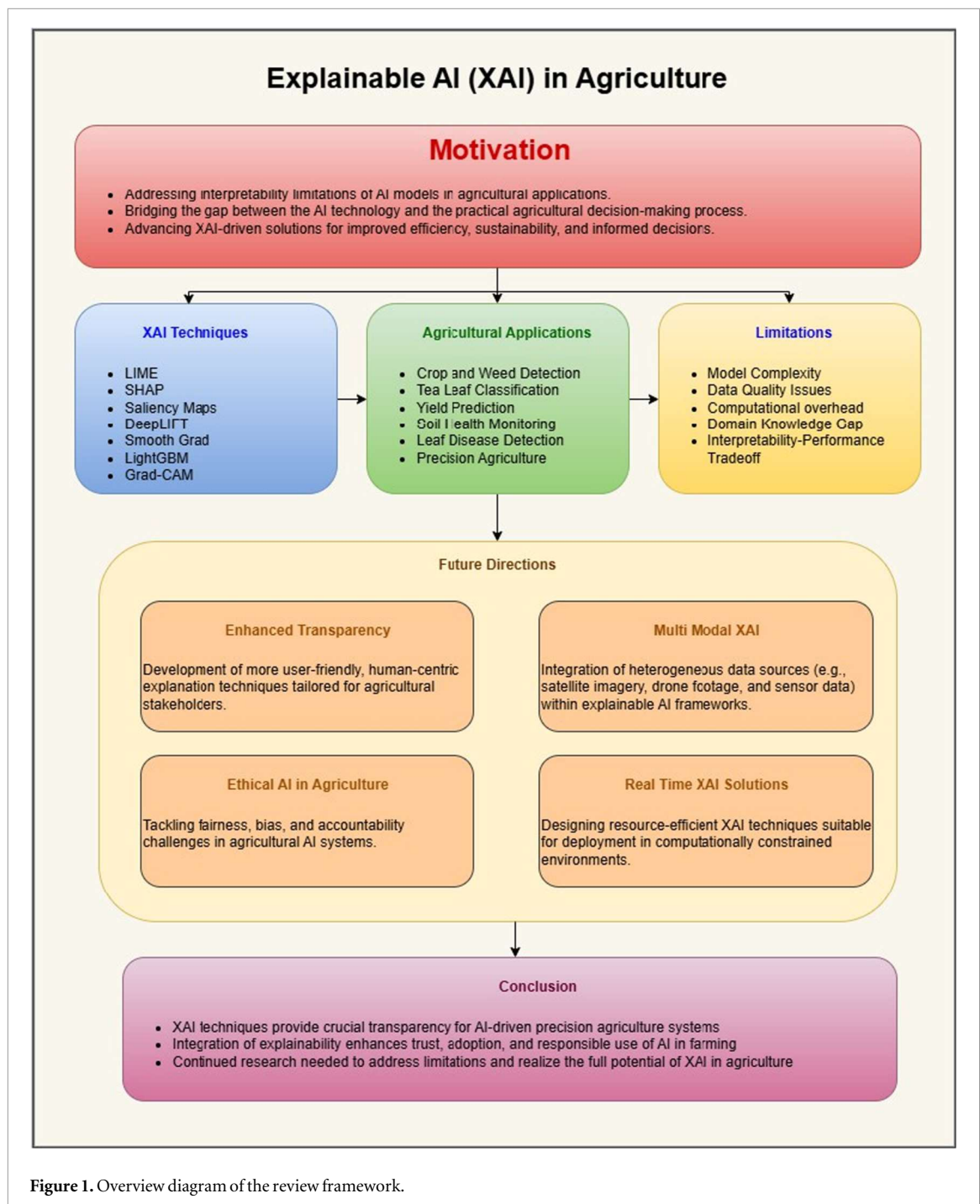
This paper reviews the research efforts in the agricultural domain that leverage XAI techniques. The review spans various key areas, including crop-weed detection, leaf disease identification, precision agriculture, yield prediction, and soil health monitoring. These domains represent critical facets of modern agriculture where XAI methods have been employed to enhance the interpretability and reliability of AI-driven solutions. This review paper aims to consolidate existing research, identify gaps, and provide insights into future directions for integrating XAI into agricultural AI systems.

Figure 3 showcases the evolution of XAI in agriculture and figure 4 depicts the list of frequently used keywords from the pertinent literature.

2. Background

2.1. AI in agriculture: evolution and applications

The definition of Artificial Intelligence has evolved over time, with classifications generally falling into three domains: rational thinking, rational behavior, and human-like thinking [13]. This conceptual foundation dates back to the 1950s when Alan Turing developed his famous Turing Test to evaluate a machine's reasoning capabilities [14]. At its core, AI consists of computational programs that process inputs and generate meaningful outputs. Its applications extend across numerous fields, including intelligent information retrieval, expert systems, robotics, automated programming, and perception-based challenges [15]. In agriculture specifically, AI is transforming traditional practices throughout the entire cultivation cycle. Machine learning techniques are used for soil assessment, crop management, decision support, aerial monitoring, automation, plant health, and weed control. AI-assisted machinery improves yield and monitoring, while predictive



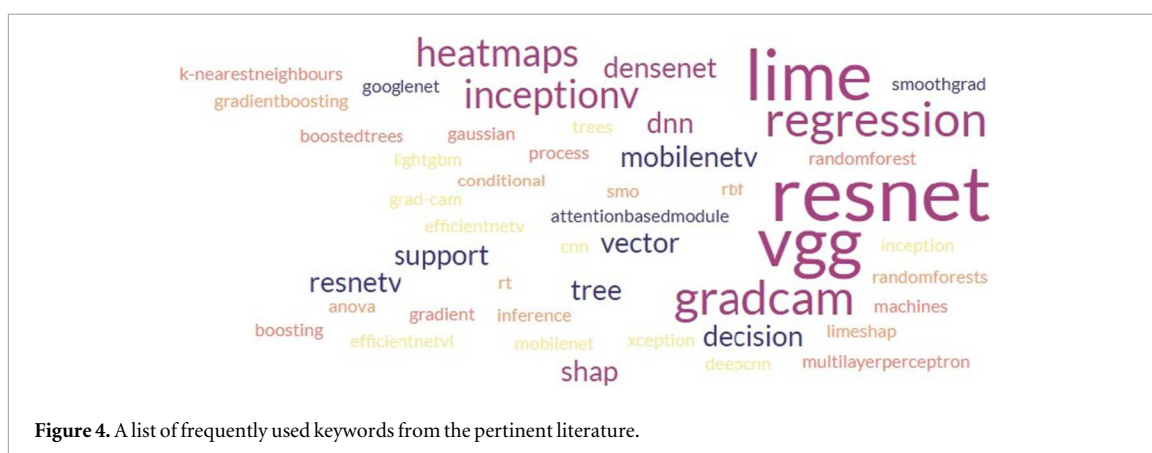
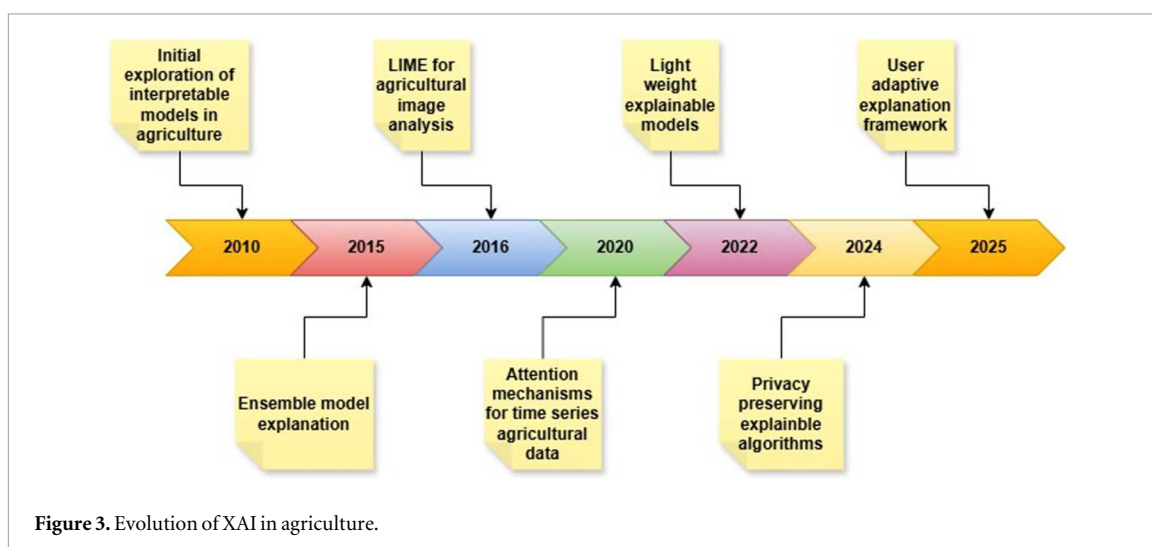
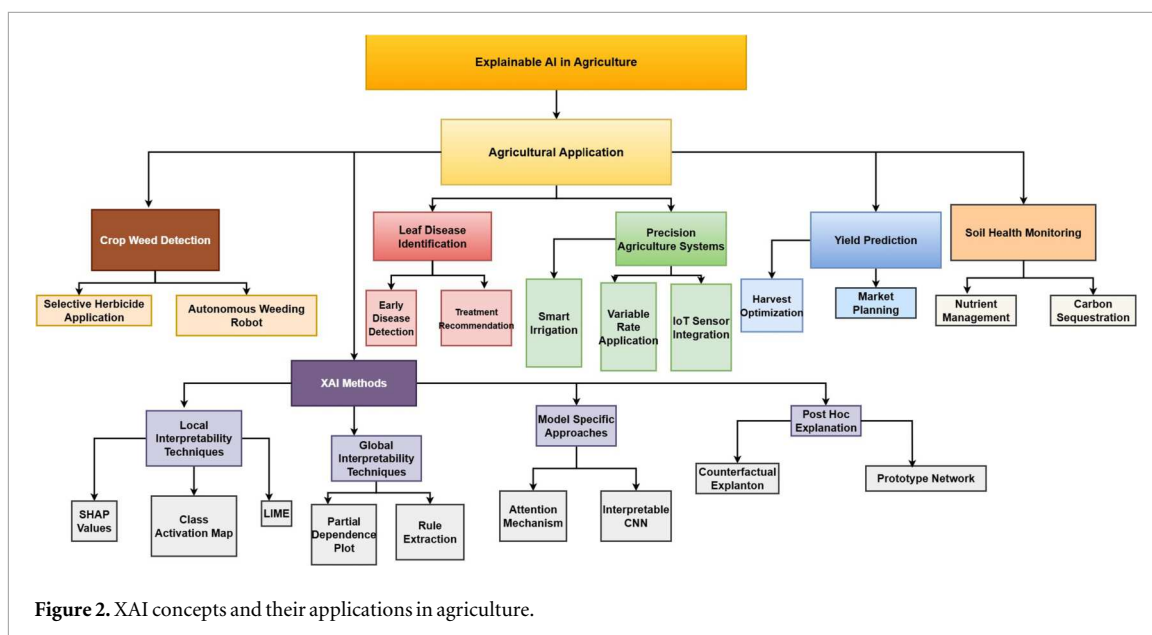
analytics aid farmers in pest control, water management, and crop rotation. Aerial monitoring and automation are also used, with AI systems detecting leaf diseases and managing unwanted vegetation.

The technological approaches powering these applications include Artificial Neural Networks, genetic algorithms, and Deep Neural Networks. While these methods offer high performance with reduced trial- and-error, they typically require substantial datasets and human expertise to function effectively [16, 17].

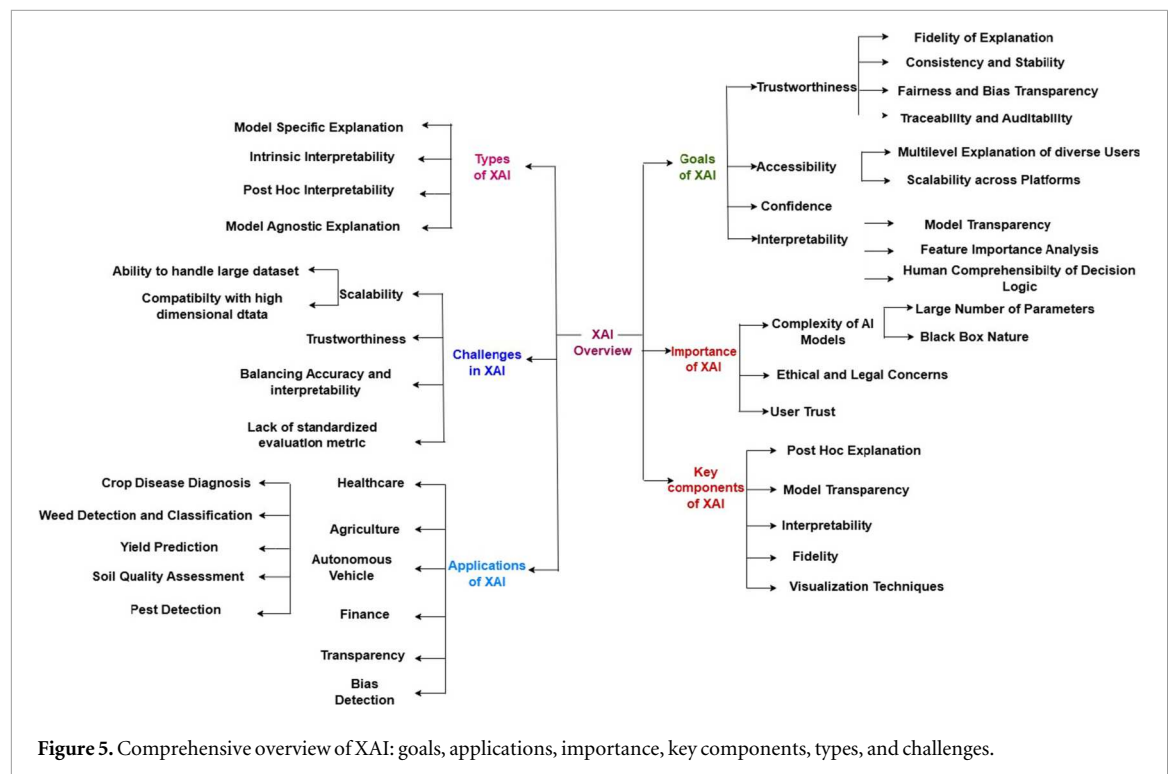
2.2. Challenges in applying ai in agriculture

Expert systems are crucial for agricultural management, but their widespread adoption in commercial agriculture is limited. Despite AI advancements, challenges persist in accuracy and response time. Real-time AI systems require processing large data volumes and identifying significant changes, which can be complex and resource-intensive.

Issues faced in the application of AI in agriculture:



- AI in agriculture presents challenges, including outdated traditional farming methods, lack of standardization, interoperability constraints, cost efficiency barriers, and demand of digital literacy among farmers. Implementation requires continuous upskilling and adaptation to evolving technological frameworks and algorithmic advancements [18].



- Machine learning models, often referred to as 'black boxes,' lack transparency and interpretability, causing a trust deficit among decision-makers in high-stakes applications like agriculture, healthcare, and finance, and hindering their adoption without additional interpretability or XAI techniques [19].
- Interdisciplinary collaboration between experts and producers is crucial for agricultural expert systems, but high costs and limited data accessibility, especially in rural areas, hinder model training and applicability [20].
- Government regulations, inadequate infrastructure, limited computational resources, and a shortage of skilled labor are significant obstacles to the adoption of advanced technologies in agriculture [21].

3. XAI techniques

XAI focuses on improving the transparency and accountability of AI by creating methods that make AI systems easier for users to understand, trust, and manage [22, 23]. In simpler terms, XAI explains how AI models make decisions by identifying the features that influence their predictions. For example, in plant disease detection, XAI can highlight the specific areas of a leaf image that the model analyzes, helping users verify that the system is focusing on the correct parts of the plant.

3.1. Overview of XAI

Deep Neural Networks (DNNs) are sophisticated artificial intelligence models that improve transparency in high-stakes situations like precision medicine and driverless cars [24]. They have complex black-box models and strong learning algorithms. Through strong learning performance and explainable models, XAI principles seek to make AI comprehensible and trustworthy for people, guaranteeing accuracy and dependability across a range of industries, including agriculture [25]. A general overview of XAI is shown in figure 5.

3.2. Goals of XAI

XAI aims to enhance transparency, interpretability, and reliability of AI systems by providing clear explanations, ensuring consistency, and promoting error identification. It also focuses on fairness, bias detection, and ethical compliance.

- Trustworthiness:** The XAI model aims to ensure trustworthiness, ensuring that the model will behave as planned when faced with an obstacle. However, not every reliable model can be explained by itself, and

reliability is not always easy to measure. It is challenging to ascertain whether trust is an explainable model's exclusive goal because the relationship between explainability and trust is nonreciprocal [25].

- **Fairness and Bias Detection:** Explainability guarantees impartiality and moral evaluation in ML models. It makes relational impacts easily observable, facilitating ethical examination. XAI seeks to draw attention to bias in data exposure. As algorithms and models become more widely used, explainability ought to stop unethical exploitation of the results [25].
- **Transparency:** Effective configuration and real-world use of AI systems depend heavily on transparency. It reviews interpretable systems for fairness and privacy protection, manages AI, and explains model judgments. Transparency is crucial to understanding how AI makes decisions, assessments, and forecasts, as it encompasses data and models. More transparency, meanwhile, could contradict other qualities and undermine the effectiveness, equity, and reliability of AI systems [33].
- **Interpretability:** Interpretability in XAI is the ability to clearly understand and explain a model's decisions, such as showing which leaf features led to a plant disease prediction in agriculture. Or in other words the capacity to communicate a model's output its cause-and-effect relationship, to a human being is correlated with interpretability [34]. Human-centered and objective methodologies assess interpretability, with particular attention paid to sparsity, simulatability, faithfulness, interaction intensity, and sensitivity to perturbations. Establishing best practices for assessing interpretation techniques and their justifications including human-computer interaction presents a difficulty [35].

3.3. Interpretable models versus explainable models

- (1) **Explainable models:** A post-hoc knowledge of the model's decision-making or prediction process is made possible by post hoc analysis using additional tools and careful study by subject-matter experts.
- (2) **Interpretable models:** The ante-hoc model design and domain expertise allow experts in the field to swiftly and easily comprehend the model's prediction or decision-making process [36].

To comprehend how intelligent systems make decisions, interpretability, and explainability are frequently used synonymously in literature. Explainability is connected to the internal logic and mechanics of a learning system, whereas interpretability is concerned with describing or expressing something in a form that is understandable to humans. While both are crucial for AI systems, interpretability is what gives humans meaningful descriptions. Both interpretability and completeness are included in explainability; completeness is defined as a precise description of how a machine learning model comes to its conclusions. Explainable models are always interpretable, however, not all interpretable models are explainable because of their incompleteness [37].

Table 1. Analysis of different XAI techniques in the agricultural domain.

Citation	Aim of the paper	Models or Algorithms	Dataset	Limitations
[26]	Explores the interpretability of deep learning models in agricultural classification tasks.	VGG GoogLeNet ResNet ResNet model + attention based module SmoothGrad LIME GradCAM	The Plant Village dataset was expanded with various fruit leaf images from Google and the AI Challenger 2018 dataset.	The paper lacks a discussion on the generalizability of findings to other agricultural datasets or classification tasks and does not address potential biases or limitations in the fruit leaves dataset used for experiments.
[27]	A hybrid model combining transfer learning and a basic CNN was created to identify sunflower diseases.	VGG19 CNN LIME	Public dataset	The research's dataset may not accurately represent the diverse disease presentations in sunflowers, potentially limiting its generalizability to real-world scenarios.
[28]	Proposes an XAI-based solution for urban vegetation mapping.	SHAP DNN	Aerial imagery of Roof Segmentation (AIRS) dataset	The DNN algorithm may lose feature-to-feature association, the kernel SHAP approach is slow for large datasets, and more investigation is needed to compare and assess current interpretation approaches.
[29]	The paper presents a method using explainable deep neural networks and a multi-layer attention procedure to classify weeds in agricultural imagery.	DNN0 ResNet5	The model is trained and evaluated on two agricultural datasets: the Plant Seedlings Dataset (PSD) and the Open Plant Phenotyping Dataset (OPPD).	The proposed model may not be robust when applied to different agricultural datasets, as it showed lower accuracy in cross-dataset evaluations.
[30]	Explores XAI in deep learning models for smart agriculture, evaluating model accuracy and using Grad-CAM for interpretation.	MobileNet ResNet50 VGG16 EfficientNetV2 DenseNet201 GradCAM approach	VegNet dataset	Not Applicable
[31]	The paper presents wearable sensing systems for agricultural workers, capturing real-time acceleration and angular velocity data. The Explainable Artificial Intelligence system was verified and tested, obtaining promising data.	SHAP LIME LightGBM	Not Mentioned	The limitations of this paper include the lack of concrete user suggestions, limited sample size, lack of discussion on method limitations, absence of external validation, and limited discussion on the limitations of wearable sensing systems.
[32]	Proposes Grad-CAM, a visual explanation technique, to enhance the interpretability of tea leaf classification models.	VGG16 InceptionV3 GradCAM	The tea leaf dataset used in this paper was manually collected from the Kangra tea garden in Palampur, India	The paper lacks details on deep learning models, pre-trained backbone models, dataset details, and performance metrics for tea leaf classification, limiting the reproducibility and generalizability of results, and potentially affecting the validity and applicability of the findings.

3.4. Common XAI techniques

Table 2 provides the definitions of various XAI approaches and comparisons between them, and table 3 categorizes XAI approaches based on their interpretability properties, computational costs, open-source tools, and agricultural data types, illustrating both theoretical distinctions and practical deployment. SHAP and SmoothGrad are computationally intensive, LightGBM has a low computational cost, and Grad-CAM is moderate. Grad-CAM and SmoothGrad are well-suited for image-based tasks, while SHAP, LIME, and LightGBM are ideal for tabular data. SHAP, LIME, and Grad-CAM enhance interpretability in crop–weed classification, whereas LightGBM and SHAP are particularly effective for crop recommender systems, offering transparency and feature relevance insights. XAI adoption in agricultural AI systems depends on model type, data modality, research goals, and practical constraints. Vision-based tasks often use model-specific visualization methods like Grad-CAM, while sensor and tabular studies use model-agnostic approaches like SHAP and LIME. Technique selection in agricultural applications is influenced by end-user preferences, computational efficiency, scalability, and interpretability-usability trade-offs.

3.5. Evaluation metric for XAI

- **Sensitivity:** Sensitivity analysis is an essential method in XAI that evaluates modifications to machine learning models such as Grad-CAM, SHAP, and LIME, assisting researchers in comprehending their robustness and dependability.
- **Faithfulness Correlation and Faithfulness Estimate:** With a high correlation signifying accurate identification and a low correlation signifying underfitting or overfitting, the statistic assesses how well XAI algorithms identify patterns and relationships in training data.
- **Monotonicity Metric:** The Monotonicity Metric is a tool that measures model performance progressively and generates contrastive explanations using monotonic attribute functions.
- **Selectivity:** In XAI models, selectivity is a statistic that evaluates the decline in the prediction function following the removal of highly attributed features.
- **Sensitivity:** Sensitivity-N determines the relationship between the target output variation and the attribution sum.
- **Infidelity:** The estimated mean square error between attribution and input perturbation and the model output difference following a large perturbation is measured by infidelity [42].

3.6. Taxonomy of XAI Techniques

- **Techniques for White Box and Black Box Model:** White box models are transparent and easy to understand [43] after training but struggle with larger datasets with intricate connections [44]. Black box models, like decision trees and Neural Networks, have intricate topologies and are challenging to understand due to their ambiguous function in the prediction model [45]. Both models have limitations regarding complexity and their ability to describe complex datasets.
- **Model-Specific Techniques and Model-Agnostic Techniques:** Model-specific techniques are explanation methods designed for a particular type of model because they rely on its internal structure. For example, Grad-CAM is used with convolutional neural networks (CNNs) to highlight which parts of a plant image a model focuses on when detecting diseases, helping farmers see the affected areas clearly. On the other hand, model-agnostic techniques can work with any model because they only use inputs and outputs without needing to know how the model works inside. For example, SHAP has been applied in agriculture to understand which factors, like rainfall, soil health, or fertilizer use, are most important in predicting crop yield. This way, model-specific techniques are ideal for tasks like image-based disease detection, while model-agnostic methods are better for analyzing patterns in agricultural data.
- **Global Interpretation and Local Interpretation:** Decision-making processes are grouped into local interpretation and global interpretation. Defining variables, dependencies, and interactions within a model requires a deeper analysis called global interpretation. Local interpretation focuses on the immediate vicinity of a data point, analyzing particular predictions and decisions. Techniques like Shapley value and LIME can help comprehend contextual relevance [44].

Table 2. Comparison Of XAI approaches in agriculture.

XAI Approach	Definition	Application	Visualized As	Advantages	Limitations	Factors Affecting Complexity
Saliency Map	AI tools that highlight areas influencing decisions by tracking changes in images [38].	Crop and weed classification, plant disease detection, fruit ripeness detecting, pest and disease detection	Heatmaps	Model understanding, debugging	May be influenced by the algorithm, may not always reflect human perception	Image size, feature extraction method
LightGBM	Microsoft's high- performance, open- source gradient boosting framework.	Crop yield prediction, crop recommender system, soil health monitoring, pest and disease prediction	Series of deci- sion trees	Faster training, feature impor- tance	Produces complex trees	Data characteristics, hyperpara- meters, learning rate, optim- ization techniques
SHAP	A game-theory-based approach that explains model outputs by treating features as players [39].	Weed detection, yield prediction, crop and weed classification, crop recommender system, soil health monitoring	Summary plot, water- fall plot, force plot	Model agnostic, global and local explanation	Computationally expensive for large datasets	Model complexity, number of features
Grad-CAM	Uses gradients to create heatmaps highlighting important regions in an image [40].	Crop disease detection, fruit ripeness classification, pest and disease detection	Heatmaps	Versatility, visual interpretation	May not always provide accurate explanations for complex models	Dataset characteristics, model- specific factors
SmoothGrad	Adds noise to input images and averages saliency maps for better interpretability.	Crop weed classification, leaf disease detection, pest detection, fruit ripeness detection	Heatmaps	Noise reduction, smoother explanations	Sensitivity to model complexity	Number of samples, model com- plexity, input dimensionality
LIME	Model-agnostic method that explains ML decisions using per- turbed samples [41].	Livestock health monitoring, crop and weed classification, soil health monitoring, pest and disease detection	Superpixel highlighting (image), bar charts (text)	Versatility, flexibility	Local interpretability, linear assumption	Perturbation strategy, feature selection

Table 3. Different XAI approaches and their characteristics.

XAI techniques	Model specific	Model agnostic	Global	Local	Intrinsic	Post-Hoc	Explanation type	Open-source tools libraries	Agricultural application or data type
LIME (Local Interpretable Model-agnostic Explanations)	✗	✓	✗	✓	✗	✓	Feature Attribution	SHAP (Python), Captum (PyTorch)	Crop recommendation systems (tabular + sensor data)
SHAP (Shapley Additive Explanations)	✗	✓	✓	✓	✗	✓	Feature Attribution	LIME (Python)	Crop yield prediction (tabular features like soil, rainfall, fertilizer)
Saliency Map	✓	✗	✗	✓	✗	✓	Visualization	Captum (PyTorch), tf-explain	Weed detection in UAV images
Decision Tree	✓	✗	✗	✓	✗	✓	Feature attribution	scikit-learn	Farm management decision-making (tabular)
Counterfactual Explanation	✗	✓	✗	✓	✗	✓	Example-Based	Alibi (Python), DiCE (Python)	Disease diagnosis (image or tabular)
Feature Importance	✓	✗	✓	✓	✓	✗	Feature Attribution	Scikit-learn, XGBoost, LightGBM	Yield prediction (tabular)
Attention Mechanism	✓	✗	✗	✓	✓	✗	Feature Attribution	HuggingFace Transformers, Keras Attention layers	Weed detection (image)
GradCAM (Gradient-weighted Class Activation Mapping)	✓	✗	✗	✓	✗	✓	Visualization	tf-explain (Tensor Flow), Captum (PyTorch), Keras-vis	Leaf disease (image)

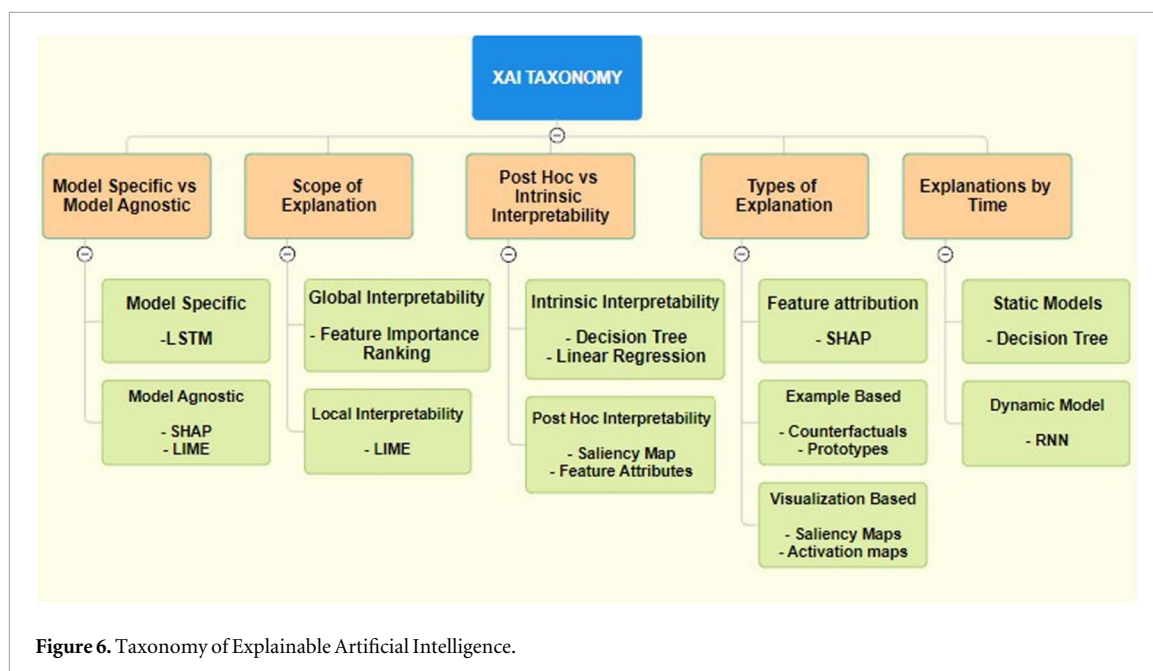


Figure 6 depicts XAI taxonomy.

4. Applications of XAI in agriculture

Tables 1, 4, and 5 provide a detailed overview of various XAI techniques, tools, and frameworks applied in the agricultural domain, along with their corresponding limitations and the datasets used.

From this analysis, it is evident that XAI has primarily been applied to classification tasks in agriculture, with additional use cases in urban vegetation mapping, plant phenotyping, and crop recommendation systems. Among the various methods, LIME and Grad-CAM emerge as the most frequently used techniques, followed by SHAP, LightGBM, and SmoothGrad.

However, several limitations have been identified. A key concern is the limited generalizability of findings to other agricultural datasets, as many studies fail to address potential biases present in their datasets. Furthermore, the real-world applicability of these models in practical farming scenarios remains largely unexplored. Another pressing challenge is the need for large, well-annotated datasets to ensure the robustness and scalability of AI models. Additionally, while SHAP has proven effective for interpretability, it is often computationally slow on large datasets, underlining the importance of developing more efficient techniques.

Recent advancements have also explored cross-modal approaches to better process and interpret agricultural data. For example, the ITF-WPI cross-modal feature fusion model, which integrates CoTN and ODLS, uses Pyramid Squeezed Attention to enhance a 4-stage network and leverages Transformer-based contextual feature extraction. Compared to similar models, the ODLS network has demonstrated strong text feature acquisition with a 30% reduction in MACCs, illustrating its efficiency and potential for practical applications such as wolfberry pest management and other agricultural use cases [46].

4.1. Crop disease detection

Crop disease detection is being transformed by the integration of AI and DL addressing the rising global demand for food and the need to protect agricultural production. Recent studies demonstrate that Explainable AI (XAI) methods can achieve remarkable performance, with one approach identifying 38 plant diseases with 99.69% accuracy, precision, and recall using LIME-based visual explanations [47]. CNN architectures such as EfficientNetB0 have been successfully applied to banana plant disease detection [48], while EfficientNetV2S attained 91% accuracy in identifying grapevine downy mildew symptoms in field conditions by detecting tiny oil spots and outperforming other CNN models [49]. Other research highlights the use of convolutional networks to improve productivity and reduce crop losses, with LIME providing the most interpretable strategy [50]. Additionally, studies combining DL with XAI methods such as LIME, Grad-CAM, and Grad-CAM++ have demonstrated the ability to accurately localize diseased regions by overlaying interpretable heatmaps, thereby enhancing diagnosis, decision-making, and transparency in agricultural disease management [51].

Table 4. Analysis of different XAI techniques in agricultural domain(contd).

Citation	Aim of the paper	Models or algo-rithms	Dataset	Limitations
[57]	Ensemble method for image classification that combines a Convolutional Neural Network with a secondary classifier trained on statistical indicators.	VGG19 DenseNet InceptionV3 Xception InceptionResNetV2	Caltech birds2011, Cars 196, Cassava, Cats versus dogs, and Citrus leaves	The datasets used for validation are not clearly described, making it challenging to evaluate the generalizability of the proposed method. The paper does not address potential limitations or challenges in integrating the proposed method with existing CNN architectures or image classification systems
[58]	The Grad-CAM technique is utilized to assess the explainability of tea-leaf classification using pre-trained deep learning models.	Grad-CAM VGG16 InceptionV3 ResNet50	Tea Leaf Dataset	Need more data to further improve the model performance.
[59]	The authors analyze XAI methods in deep learning for plant phenotyping research, finding they can identify unknown features, improve current models, and clarify results, suggesting XAI methods can enhance plant phenotyping tools.	The paper does not focus on a specific model or algorithm. Instead, it provides an overview of the different approaches that have been used in the field and discusses the advantages and limitations of each approach.	Public dataset	The authors do acknowledge some limitations of deep learning-based approaches in plant phenotyping, such as the need for large labeled datasets and the potential for overfitting. They also note that some deep learning models may be too complex for practical use in the field.
[60]	The authors use Inception V3 and ResNet models on Plant Village and New Plant Disease Datasets to detect and classify crop diseases using Explainable AI tools.	ResNet model LIME Grad CAM	Plant village dataset and New plant disease dataset (created by the use of the augmentation method from the original Plant Village dataset).	The paper lacks specific limitations of proposed models, XAI techniques, and datasets used for training and prediction, and provides no insights into the scalability or generalizability of the approach.
[61]	This paper shows the application outcomes of a computer vision system that integrates LoRa and Deep Learning and is capable of effectively identifying grape leaf diseases from low-resolution photos.	MobileNetV2 ResNet50	PlantVillage, which contains images of grape crops and other crop species.	The paper lacks a detailed analysis of the system's performance in real-world scenarios, comparisons with other image transmission methods, and specific details about experiments and dataset size.
[62]	The authors utilized three pre-trained CNN models to identify 38 diseases across 14 distinct plant species.	EfficientNetV2L MobileNetV2 ResNet152V2 LIME	The dataset was collected from Kaggle.	The dataset used in this study was limited to 5000 images.

Table 4. (Continued.)

Citation	Aim of the paper	Models or algo-rithms	Dataset	Limitations
[63]	Proposes the creation of an imageprocessing system that uses CNN and Deep Learning to classify leaf illnesses.	Deep CNN model LIME	Not Mentioned	The paper lacks details on the dataset used, plant species considered , and the generalizability of the proposed models to different environmental conditions or plant varieties.
[64]	A crop recommender system is presented in this paper to help with crop recommendations based on soil nutrient levels and metrological parameters.	SHAP RBF SMO ANOVA	Kaggle	The study's potential flaw is its inability to validate its method with real farmers or agricultural specialists, limiting classifications to seven features in the dataset.

Table 5. Analysis of different XAI techniques in agricultural domain (contd.).

Citation	Aim of the paper	Models or Algo-rithms	Dataset	Limitations
[73]	This paper proposes an XAI framework (OAK4XAI) to address the challenge of interpreting knowl- edge in digital agriculture. The framework considers data analysis and semantics, resulting in an Agriculture Computing Ontology (AgriComO) that provides a well- designed structure for understanding knowledge in agriculture and computing domains.	OAK4XAI model AgriComO	Not Mentioned	Doesn't focus on the reasoning module and explanation interface for multiple user groups.
[12]	Illustrated the potential applications of a set of interpretable machine learning technologies for the processing of agricultural data.	Linear model with AIC stepwise variable selection Decision tree (condi- tional inference tree) Random Forests Gradient Boosting LIME	Public Dataset	Ought to have concentrated on expanding the XAI approaches tool box. Neglected to address bias mitigation in agri- cultural datasets.
[74]	The study investigates the factors influencing the production of the flagship crops in 32 Mexican states.	RT Boosted Trees Support Vector Machines (SVM) Gaussian Process Regression (GPR)	Mexican National Water Commission (Comi- sio'n Nacional del Agua, CONAGUA) Agro-Food and Fisheries Information Service (Servicio de Informacio'n Agroalimentaria y Pesquera, SIAP)	Temporal and spatial analysis are not incorporated.
[75]	Nitrogen status of wheat was estimated using machine learning ap- proaches using hyperspectral data.	Support Vector Regression Random Forest K-Nearest Neighbours Multilayer Perceptron Gradient Boosting Regression Partial Least Square Regression SHAP	Gilat Agricultural Research Center, Israel.	Although the study pinpoints critical wavelengths for estimating wheat growth stages' nitrogen status, their specificity might prevent them from being applied to other crops or environments.
[66]	Introduces a model that integrates eXplainable Artificial Intelligence (XAI) with Predictive Maintenance (PdM), offering predictive insights and explanations across four key dimensions.	LSTM XGBoost	Not Applicable	Not focused multi-modal data in-tegration and Human-in-the-Loop (HITL) systems.

For detecting agricultural diseases in Internet of Things (IoT) edge environments, RepAgrViT is introduced as a lightweight vision transformer architecture. It incorporates an efficient parameter fusion strategy, integrates CNN features with vision transformer concepts, and employs local-global attention mechanisms. Additionally, the model leverages neighborhood-based local search strategies and adaptive weighted delayed velocity optimization. Experimental findings demonstrate that RepAgrViT effectively captures discriminative disease features, making it suitable for resource-constrained agricultural monitoring systems [52].

A multi-level deep information feature fusion extraction network (DFN-PSAN) is proposed for plant disease classification in natural field scenarios. This network processes multi-level depth information and emphasizes key regions of plant disease images through pyramidal squeezed attention (PSA) and multiple convolutional layers. The proposed approach achieved strong accuracy and improved model attention transparency, demonstrating competitive performance [53].

The study also presents PPLC-Net, a deep learning model designed to enhance feature extraction and expand sample diversity using meteorological data augmentation. It addresses spatial information challenges through GAP layers, a lightweight CBAM attention mechanism, and sawtooth dilated convolution. The model achieved an F1 score of 98.442% and a recognition accuracy of 99.702%, fulfilling the criteria for precise and efficient recognition. Furthermore, the integrated CAM-based visualization method validated its interpretability [54].

To support crop disease diagnosis, the article proposes YOLO V5-CaCT, an optimized detection framework built on the lightweight YOLO V5 backbone. The network incorporates Repeated Augmentation, Focal Loss, and SmoothBCE for improved robustness and convergence. The model is optimized using simplified operators and trained through two technical pathways, reducing model size and inference time while achieving 94.24% average recognition accuracy across 59 crop disease categories [55].

4.2. Weed detection and classification

In weed detection and classification, XAI has been integrated with advanced deep learning models to improve accuracy, interpretability, and real-time applicability. A high-resolution attention architecture based on ResNet layers was proposed to enhance the localization of weed patches, enabling transparent comparisons across weed species and growth stages through self-attention maps [29]. This approach maintains accuracy from deeper layers while improving resolution through shallower ones, thereby supporting cross-dataset evaluations. Similarly, a modified YOLOv8-nano model was developed for edge device deployment, leveraging the CottonWeedDet12 dataset with 16,944 images to detect 12 cotton weed classes [56]. The incorporation of a convolutional block attention module and C3Ghost block reduced computational costs while maintaining reliability, and interpretability was further enhanced using XAI techniques such as Layer-CAM, Grad-CAM++, and Eigen-CAM, which provided visual explanations of weed detection outcomes.

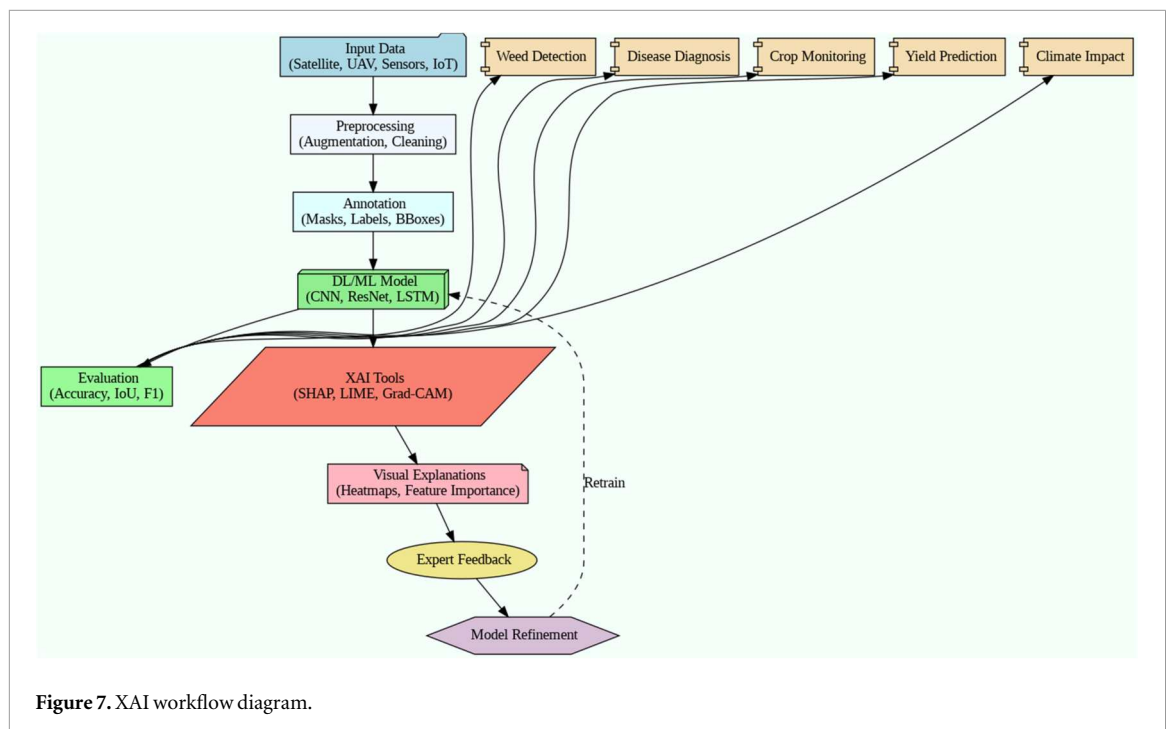
4.3. Precision agriculture

Crop recommendation systems have greatly improved in terms of performance and interpretability because of XAI approaches. A study [64] suggested a recommender model that improved decision-making transparency with an accuracy of 98.2%. The XAI-CROP framework, which was created using datasets from Indian crop farming, performed better than conventional models and provided convincing arguments for its suggestions [65]. With excellent scalability across many crops and areas, this improves food security, transparency, and resource utilization. Predictive maintenance in smart agricultural facilities has also incorporated XAI, which offers interpretable insights for data, models, and end users. This has improved ROC-AUC, accuracy, and F1-score in comparison to current methods [66].

4.4. Yield prediction

In precision agriculture, XAI techniques are increasingly being employed to enhance crop prediction and recommendation systems. A Bayesian interpretable machine learning model has been proposed for yield prediction under future climate conditions, outperforming black-box models such as MARS, SVM, neural networks, Random Forest, and XGBoost by capturing structural discontinuities and nonlinear effects of temperature and precipitation [67]. Similarly, the XAI-CROP approach, developed using Python, Pandas, Scikit-learn, and LIME, improves transparency and reliability in crop recommendation systems in America [68]. Other studies have applied explainable boosting machines for cotton yield prediction, highlighting the role of key input features [69], while satellite imagery-based models have been used to quantify input contributions, track crop growth stages, and analyze production variability with feature attribution methods [70]. These advancements demonstrate how XAI improves interpretability, decision-making, and trust in agricultural prediction systems.

Figure 7 shows XAI workflow diagram.



4.5. Soil health monitoring

XAI has also been applied to soil health monitoring, particularly for optimizing cultivation practices. For rice production, XAI-based models incorporate feedback loops and real-time updates to enhance prediction accuracy while considering climate conditions, thereby supporting sustainable operations through soil quality optimization [71]. Similarly, in the European Union, XAI-driven models have been employed to predict soil fertility by analyzing contributing factors, with results presented through intuitive visualizations that demonstrate their potential for strengthening global food security [72].

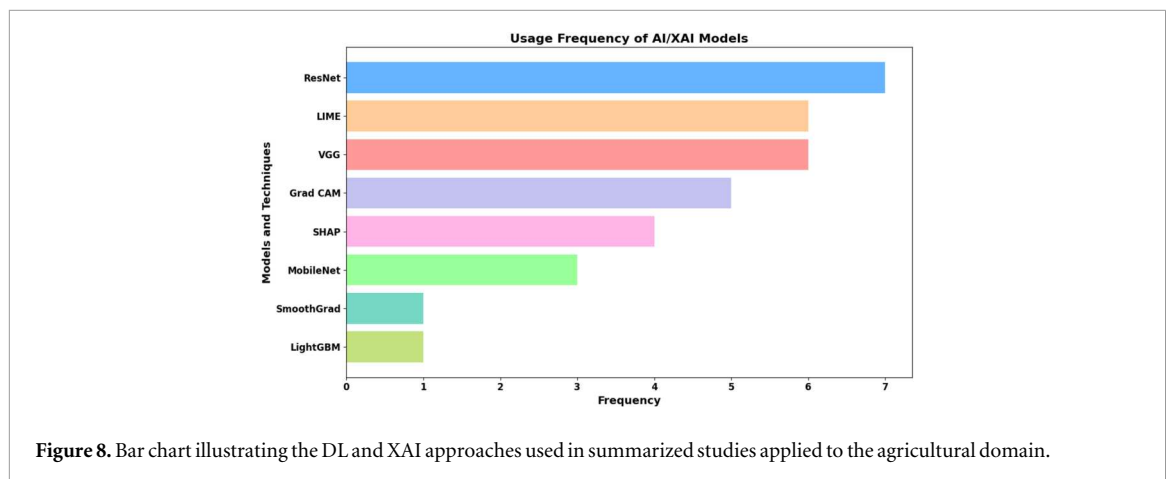
4.6. Performance metric trends in reviewed literature

It is clear from the examined literature that, depending on the task, researchers use a variety of evaluation criteria. Although accuracy is still the most frequently reported metric, several studies now incorporate precision, recall, and F1-score to more thoroughly represent performance at the class level. Furthermore, research that uses regression-oriented methodologies, such as growth modeling or yield prediction, usually reports error-based metrics like the coefficient of determination (R^2), mean squared error (MSE), and mean absolute error (MAE). Direct comparison between research is difficult due to the variation in reported indicators, but it also highlights how complex performance evaluation is in agricultural AI applications. Crucially, accuracy alone might not adequately convey a model's robustness or field-readiness, especially in situations where balancing error rates (F1-score) or reducing false negatives (recall) are crucial, or where inference time and computational efficiency have an impact on real-world deployment. Thus, implementing a complete and consistent set of measures will guarantee that AI-based solutions are assessed in accordance with the intricate requirements of actual agricultural contexts, in addition to improving comparability between research.

5. XAI across diverse fields

[76] provides a comprehensive review highlighting the widespread adoption of AI across multiple fields, including agriculture, materials science, device modeling, and healthcare. Recent research has leveraged XAI techniques such as LIME and SHAP to analyze nanowire gate-all-around ferroelectric FETs for neuromorphic computing applications. XAI has also been instrumental in enhancing diagnostic transparency in healthcare and accelerating materials discovery, underscoring its universal role in improving trust, reliability, and interpretability in AI systems.

In healthcare, medical image analysis stands out as a domain where AI has been extensively applied, with high accuracy complemented by improved model explainability and interpretability. Many studies in this area focus on distinguishing between diseased and healthy tissues using visual tools such as saliency maps [77] which introduces the explanation interfaces designed to support automated diagnostic and predictive tasks, primarily



used by physicians. Similarly [78], presents a dashboard that integrates data from sensor-equipped smart homes to detect cognitive decline in elderly individuals and alert healthcare professionals. In the field of psychiatry and mental health, [79] emphasizes the importance of interpretability and transparency, introducing the TIFU framework to make AI models more comprehensible. Here, AI is primarily applied for prediction and discovery, where explainability remains critical for ensuring trust and safety in high-stakes clinical decision-making.

Using both global and local explanations, [80] suggests an end-to-end XAI framework for deciphering and displaying anomaly detection classifications in AI models for VANET security of autonomous vehicles. As a way to illustrate the reliability and explainability of an AI-based IV-IDS (Anomaly-based In-Vehicle Intrusion Detection System), [81] created a visualization-based explanation called VisExp, highlighting feature and SHAP values, utilizing Python libraries and SHAP.

6. Discussion

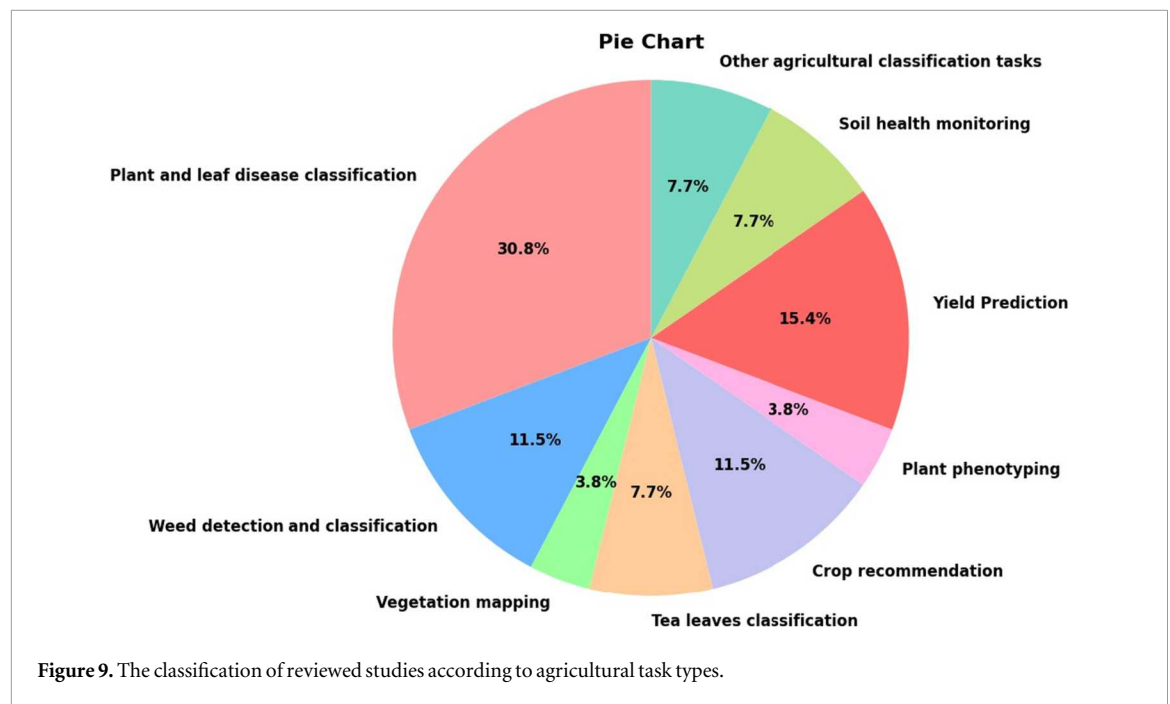
Figure 8 shows a bar chart illustrating the DL and XAI approaches employed in the summarized studies within the agricultural domain. Among the DL models, ResNet exhibits the highest usage, followed by VGG. In the XAI category, LIME is the most frequently employed method, followed by Grad-CAM, while SmoothGrad and LightGBM are comparatively less utilized. Grad-CAM is widely used due to its versatility, as it can be applied to image classification, object detection, and semantic segmentation tasks. Similarly, LIME's model-agnostic nature, along with its flexibility and adaptability, makes it a popular choice among XAI approaches compared to others.

Figure 9 illustrates the classification of reviewed studies based on different agricultural task types. The majority of agricultural tasks utilizing XAI approaches focus on plant and leaf disease classification, followed by yield prediction, while vegetation mapping and plant phenotyping are the least explored. This trend can be attributed to the abundance of image data, as large-scale datasets of diseased plant images and crop growth patterns are readily available, facilitating the application of deep learning and XAI techniques. In contrast, vegetation mapping and plant phenotyping involve complex environmental factors and large-scale spatial data, making the integration of XAI techniques more challenging.

Stakeholder needs, data modality, and model type all influence how XAI techniques are adopted in agriculture: While feature significance and SHAP are typical for tabular yield models, visual attribution tools (Grad-CAM, saliency) predominate in image-based disease/weed tasks, and new counterfactual and concept-based approaches are becoming more and more respected for their applicability and conformity to agronomic principles.

7. Challenges and limitations

Creating a fully functional white box system that produces intelligible lay-person explanations with significant user acceptability is XAI's primary issue. This requires the collaboration of ML and XAI researchers, as well as user interface, user experience, usability, and human-computer interaction researchers. The goal of XAI is to make decisions understandable and usable by human stakeholders who are not accustomed to technology. Stakeholders of adoption, accessibility, and acceptance include domain experts, regulatory agencies, top



executives, and direct users of XAI. To achieve usability and acceptance, an amalgamation of XAI and technology acceptance models should be used [82].

XAI enhances machine learning models, particularly those with abstract features. However, its effectiveness depends on various factors like task, dataset, model architecture, explanation technique, and hyperparameters [83]. Challenges include adaptive integration, consistent explanations, and interactive explanations [84]. To ensure long term use and maintainability, resource management, algorithm efficiency optimization, and balancing predicted accuracy and model interpretability are crucial.

DL models trained on extensive plant phenotyping datasets often struggle with real world conditions, where variables like fluctuating light and temperature are common. These models can be heavily influenced by factors such as temperature variations, types of germination materials, pH levels of nutrient solutions, aeration, lighting conditions, filter paper colors, and camera settings. Such biases make researchers question the generalizability and reliability of deep learning models for plant studies. Relying on these models for assessing plant varieties could lead to inaccurate results and suboptimal decision-making, posing potential risks for the future of plant research [59].

Other challenges include:

- It is crucial but difficult to protect data privacy and security, integrate many data sources (geospatial, sensor, and historical), and guarantee dependable connectivity [85].
- Changes in labour, knowledge-sharing, and benefit distribution brought about by XAI necessitate cooperative prioritization and decision-making, which stakeholders must adjust to [86].
- To protect data integrity, it is essential to address cybersecurity, user permission, and data ownership [85].
- For XAI systems to be equitable, all farmers must have access to and be able to access them. It is difficult to effectively communicate sophisticated AI results to a variety of stakeholders. Numerous researches have documented the practical and computational issues that arise when balancing model simplicity and accuracy in real-world applications [83, 85, 87–89].
- AI and XAI in agriculture face challenges like data availability and computational complexity, which may hinder widespread adoption, especially in resource-limited regions [90].

8. Future directions

The integration of Explainable Artificial Intelligence (XAI) into agricultural systems holds transformative potential for enhancing the transparency, reliability, and ethical deployment of AI-enabled decision support tools [91]. By addressing critical challenges related to interpretability, model accuracy, bias detection, and user

trust, XAI facilitates informed decision-making among key stakeholders, including farmers, agronomists, researchers, and policymakers. Unlike conventional ‘black-box’ models, XAI frameworks provide human-interpretable rationales for predictions or classifications, thereby promoting transparency and accountability in data-driven agricultural interventions.

Moreover, the synergistic integration of XAI with emerging technologies such as blockchain, the Internet of Things (IoT), and edge computing, which enables data processing directly on local devices like drones or farm sensors rather than distant servers, hold promise for real-time agricultural applications. Edge computing enhances the efficiency, security, and scalability of smart farming systems [92, 93]. For instance, XAI combined with blockchain can enable tamper-proof data auditing and traceability in supply chain logistics, while XAI-guided IoT deployments allow for real-time monitoring and autonomous decision-making with improved interpretability at the edge. These integrated systems facilitate the deployment of precision agriculture practices, optimize resource utilization, and enhance food safety monitoring through traceable and interpretable data analytics pipelines.

Despite these advancements, several technical and socio-economic challenges persist. The high maintenance costs of XAI-integrated systems, the absence of standardized automation protocols, and unresolved ethical dilemmas associated with opaque or biased AI decision-making represent significant barriers to widespread adoption [13, 94–96]. Additionally, the lack of user-friendly interfaces and limited domain-specific interpretability tools impede the accessibility of XAI for non-expert end users, particularly in resource-constrained rural settings.

Future research should prioritize the development of sustainable, context-aware, and inclusive XAI frameworks that uphold the principles of fairness, accountability, and transparency. This includes the design of lightweight and explainable models suitable for deployment in low-power environments, the integration of multi-modal sensor data for robust interpretability, and the incorporation of domain-specific knowledge into XAI algorithms. Moreover, frameworks must be evaluated not only on performance metrics such as accuracy and precision but also on their explainability, usability, and social acceptability among target user groups.

While Agriculture 5.0 envisions a highly interconnected, intelligent, and sustainable agricultural paradigm, an over-dependence on automated technologies may inadvertently marginalize traditional farming knowledge systems and impact rural employment. To ensure a balanced and ethical transition, future AI models must incorporate principles of environmental sustainability, legal transparency, and human-centered governance. This necessitates active collaboration between interdisciplinary stakeholders—including data scientists, ethicists, legal experts, policymakers, and agricultural practitioners—to co-design responsible AI ecosystems tailored to local socio-economic contexts [85], [97–102].

In summary, the responsible and strategic integration of XAI into agriculture presents a promising avenue for building resilient, inclusive, and efficient agri-food systems. When aligned with ethical frameworks and participatory development models, XAI can serve as a catalyst for equitable technological transformation in the agricultural domain.

9. Conclusion

The agricultural sector is crucial for economic growth, employing significant human capital globally. Precision agriculture helps address challenges like climate volatility, biological invasions, and resource scarcity. Ethical concerns and regulatory mandates have sparked interest in model interpretability and explainability frameworks, with research focusing on decentralized marketplace architectures, blockchain-enabled provenance systems, and explainable AI implementations. XAI methodologies extend beyond weed differentiation and disease diagnostics to predictive analytics for cultivar selection, optimizing resource allocation, enhancing food security, and maximizing agricultural production. They integrate post-hoc interpretability techniques like Grad-CAM, LIME, and SHAP, improving user interface accessibility, algorithmic transparency, and stakeholder confidence among agricultural practitioners. The use of interpretable artificial intelligence (XAI) in agriculture has shown significant progress, with traditional machine learning and deep learning architectures being the most commonly used. XAI-enabled precision agriculture systems can reduce water consumption by 15%–25% while improving crop yields. However, regional disparities exist, with developed economies implementing XAI more quickly than emerging markets. The regulatory landscape for AI transparency in agriculture is evolving, emphasizing model auditability and stakeholder accessibility. Implementation barriers include technical challenges in edge computing deployment and underdeveloped knowledge transfer mechanisms for translating complex model explanations into actionable insights. The review emphasizes the critical necessity for collaborative ecosystems encompassing farmers, agricultural stakeholders, and computational scientists to facilitate the seamless integration of model-agnostic interpretability techniques with established agricultural methodologies. Future research priorities

should address cross- domain knowledge transfer from adjacent fields, develop standardized validation frameworks for agricultural XAI, and create specialized explainability approaches for handling multimodal agricultural data streams including satellite imagery, IoT sensor networks, and genomic information. This systematic assessment contributes substantively to the evolving discourse surrounding XAI applications in agriculture and provides empirically grounded insights regarding its potential benefits for sustainable agricultural intensification and resilient food systems in an increasingly resource- constrained global environment.

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Data availability statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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